

REDMOND, OR, USA

WMO: 726920

Lat: 44.256N

Lon: 121.139W

Elev: 928

StdP: 90.67

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24230

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-14.4	-10.9	-17.9	0.9	-12.4	-14.6	1.2	-7.6	11.1	7.0	9.7	6.4	1.9	330	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	34.2	16.3	32.6	15.9	31.0	15.4	17.6	31.3	16.8	30.0	15.9	28.6	4.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	10.3	19.5	11.7	9.6	19.3	10.7	8.9	19.2	52.8	31.1	50.0	29.7	47.5	28.6	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	DB	-19.9	37.5	4.6	1.7	-23.2	38.8	-25.9	39.8	-28.4	40.8	-31.8	42.1	(4)
(5)			WB	-20.3	19.1	4.5	0.9	-23.5	19.8	-26.1	20.4	-28.7	20.9	-31.9	21.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	1.1	1.9	4.6	6.9	11.3	15.0	19.6	18.9	14.5	8.5	3.7	0.1		
	DBStd	7.83	4.84	4.43	3.67	3.72	4.13	3.93	3.47	3.32	3.95	4.03	4.96	5.03		
	HDD10.0	1402	276	227	170	107	33	5	0	1	7	76	194	308		
	HDD18.3	3587	533	459	424	342	221	114	27	33	125	304	438	566		
	CDD10.0	1002	1	1	3	14	73	154	299	278	143	30	6	1		
	CDD18.3	147	0	0	0	0	3	14	67	52	11	0	0	0		
	CDH23.3	4010	0	0	1	22	153	466	1586	1276	475	32	0	0		
	CDH26.7	1742	0	0	0	2	41	167	772	582	173	4	0	0		
(14) Wind	WSAvg	2.8	2.6	2.8	3.1	3.1	3.1	3.2	2.9	2.8	2.6	2.6	2.7	2.6		
(15) Precipitation	PrecAvg	219	27	16	19	15	21	18	11	12	10	14	29	28		
	PrecMax	315	78	60	49	44	59	98	95	64	32	48	80	139		
	PrecMin	128	3	0	0	0	1	0	0	0	0	0	3	0		
	PrecStd	43	16	13	13	12	16	20	17	15	9	10	21	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	17.5	21.3	26.2	30.8	34.1	37.1	36.5	33.5	26.7	20.5	14.1		
		MCWB	7.8	8.2	10.1	12.7	14.9	17.0	17.8	16.5	15.6	12.8	10.7	7.6		
	2%	DB	12.3	13.7	17.8	22.5	27.4	30.9	34.8	33.9	31.1	23.5	16.9	11.4		
		MCWB	7.0	7.2	8.4	11.3	13.8	15.8	16.6	16.0	14.8	11.4	9.7	6.7		
	5%	DB	10.0	11.3	15.1	19.1	24.4	28.3	33.0	32.2	28.5	20.7	13.9	9.3		
		MCWB	5.5	6.0	7.2	9.6	12.5	14.6	16.1	15.7	14.2	10.6	8.0	5.7		
	10%	DB	8.0	9.0	12.6	16.2	21.9	26.0	31.3	30.2	26.1	17.7	11.6	7.5		
		MCWB	4.4	4.6	6.6	8.2	11.6	13.3	15.5	15.1	13.2	9.6	6.8	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.6	11.2	13.6	16.1	18.0	19.2	18.5	17.2	14.1	12.3	9.2		
		MCDB	13.5	16.4	17.9	23.9	27.7	31.2	34.2	32.2	29.9	22.9	17.9	12.7		
	2%	WB	7.2	7.8	9.7	11.8	14.6	16.3	17.8	17.4	15.7	12.5	10.1	7.3		
		MCDB	11.5	13.1	14.4	21.6	24.8	28.9	32.0	30.9	27.9	21.0	15.7	10.6		
	5%	WB	5.8	6.4	8.4	10.2	13.4	15.2	16.9	16.5	14.8	11.3	8.6	5.8		
		MCDB	9.4	10.7	14.7	18.3	22.8	26.7	30.6	29.5	26.6	19.1	13.7	9.1		
	10%	WB	4.7	4.9	6.9	8.8	12.1	14.0	16.1	15.6	13.7	10.1	7.0	4.5		
		MCDB	7.9	8.6	12.7	15.4	20.4	24.4	29.2	28.1	25.0	17.1	11.0	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.5	14.1	15.5	16.4	18.1	21.0	20.7	19.9	16.5	12.1	10.0		
		MCDBR	12.9	15.0	18.9	22.0	21.7	22.2	23.2	23.3	24.1	21.8	15.6	11.8		
	5% WB	MCWBR	8.2	9.4	11.0	11.9	10.1	9.5	9.1	9.5	11.0	11.7	9.4	7.9		
		MCWBR	11.6	13.6	17.4	20.0	19.3	20.8	21.4	21.5	22.0	18.5	14.4	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.288	0.289	0.305	0.323	0.317	0.331	0.321	0.304	0.301	0.279	0.280		
	taud		2.542	2.559	2.578	2.529	2.495	2.539	2.507	2.551	2.571	2.570	2.593	2.548		
	Ebn at Noon		857	918	959	962	946	949	932	934	929	888	859	826		
	Edh at Noon		66	77	86	98	105	101	103	95	86	75	62	60		
(44) All-Sky Solar Radiation	RadAvg		1.56	2.46	3.65	5.13	6.26	7.27	7.79	6.68	5.09	3.17	1.80	1.27		
	RadStd		0.18	0.30	0.31	0.44	0.47	0.39	0.32	0.36	0.28	0.26	0.18	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	+37

Nomenclature: See separate page